

Old Film, Live Lamp

Object Custody for Long-Horizon AI Memory

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Abstract

Long-horizon AI work is often described as a memory problem. The usual task is to store more prior material, compress it, retrieve it, and place it back into the working context. This paper argues that the missing unit is not storage. The missing unit is custody. A long-running system must decide what remembered material is allowed to do.

The proposed model is **Old Film, Live Lamp**. Old film names transcripts, files, generated drafts, source packs, prior answers, reports, and rejected routes. The live lamp names the small active-state surface that may steer the next move. A rolling state protocol moves the lamp through time without letting the whole archive drive. The result is a custody model: old material may supply evidence, search paths, and contrast cases, but the current object, current user terms, and current source gate decide action.

The paper is grounded in a live archive case. The working corpus grew into a multi-million-token file world with nested archives, duplicates, reports, paper drafts, source ledgers, and generated PDFs. The usable method was not to pour that whole world into active context. The usable method was to preserve route, role, source path, version status, and live-object fit. This paper gives definitions, a custody protocol, an audit table, and testable metrics for long-horizon AI memory systems.

Keywords

long-context AI; AI memory; retrieval-augmented generation; provenance; active state; archive governance; source custody; human-AI scholarship

1. Problem

A long AI session can fail even when the needed text exists somewhere in the archive.

Common failure modes include:

1. **Storage without role.** A source is present, but the system does not know whether it is current, archived, rejected, draft, reference, or counterexample.
2. **Summary as substitute.** A summary returns as if it were the object.

3. **Path without object.** The system asks for a filename when the user remembers a project by shape, creature, task, failure, or phrase.
4. **Object without source.** A live answer fits the object but lacks a source edge.
5. **Old mass as driver.** Large prior context overpowers current user terms.
6. **Tiny live cue ignored.** A recent instruction is demoted because it is small compared with the archive.
7. **Plausible neighbor promoted.** A nearby source family is treated as proof.

The core question is therefore not only:

Can the system remember?

It is:

What role may remembered material play in this turn?

2. Terms

2.1 Object

The object is the thing under work. It may be a paper, claim, bug, method, prompt, diagram, route, decision, or source family. The object is not the person, not the mood around the task, and not the largest pile of related text.

2.2 Old Film

Old film is archive material. It includes prior chat, file extracts, draft papers, reports, ZIP contents, source maps, test logs, and generated artifacts. Old film may guide search and support a claim, but it does not steer by mass.

2.3 Live Lamp

The live lamp is active state. It contains the current user objective, current constraints, current source target, current permission, current non-goals, and current next action.

2.4 Rolling State

Rolling state is the procedure that updates the live lamp. It keeps the lamp small enough to act while preserving enough source route to return to the archive when needed.

2.5 Source Nearness

Source nearness is the ability to reach the right source family from the shape of the object. It is not source truth. A nearby source can be a lead, not proof.

2.6 Custody

Custody is role control over memory. Each memory item receives a role before it is used:

```
current steering
source evidence
search lead
draft artifact
rejected route
counterexample
unknown
```

3. Model

Old Film, Live Lamp separates archive capacity from steering authority.

```
OLD FILM:
archive, evidence, search path, prior output

LIVE LAMP:
current object, current terms, current permission, current next move

ROLLING STATE:
small update surface that carries the lamp forward

SOURCE GATE:
path, signal, status, and version check before source claims

OBJECT GATE:
object-in compared with object-out before action
```

The design rule:

```
Memory is not steering by default.
```

A memory item may become steering only when the live lamp grants that role. Otherwise it stays as evidence, search path, contrast case, or dormant archive.

4. Relation to Existing Work

MemGPT frames long interaction as virtual context management across memory tiers [1]. Retrieval-augmented generation supplies external memory through retrieval [2,3]. Studies of long-context use show that a large context window does not ensure usable retrieval across positions [4,5]. ReAct links reasoning traces to action steps [6]. W3C PROV gives vocabulary for provenance relations among entities, activities, and agents [7]. Datasheets and Model Cards show how documentation artifacts can state intended use, provenance, and constraints [8,9].

Old Film, Live Lamp uses those neighbors but asks a different question. It asks what a remembered item is allowed to do. Retrieval answers where text may be found. Provenance answers where a source edge came

from. Documentation answers how an artifact should be read. Custody adds role control: whether an old item may steer, support, route, remain dormant, or be refused as driver.

5. Case Basis

The case corpus behind this paper includes nested ZIPs, duplicate files, paper drafts, source maps, rendered PDFs, manifests, and intake reports. The case showed three recurring facts.

First, archive size and active steering are different variables. A corpus can contain millions of estimated tokens while the live steering layer remains small.

Second, duplicate and near-duplicate material can increase apparent agreement without adding independent evidence. Version and path status must be tracked.

Third, object memory and path memory diverge. The user may remember “the old-film repair” or “source-near is not source-true” while the archive stores paths, filenames, and versions. A useful assistant converts object cues into source routes, then marks the route status before making a claim.

The case is not a benchmark. It is a design specimen. Its value is the failure pressure it exposes: a long-memory system must preserve object, role, route, version, and source edge under compression.

6. Custody Protocol

The protocol has seven steps.

Step 1: Name the live object

OBJECT:
What is being handled now?

The answer must be specific enough to compare object-in with object-out.

Step 2: Mark user terms

TERMS:
What constraints, vetoes, format needs, and non-goals govern this turn?

Recent terms outrank archive habit.

Step 3: Separate old film from lamp

OLD:
What archived material may help?

LAMP:

What is allowed to steer?

The old archive may suggest where to look. It does not become current instruction without permission.

Step 4: Assign source role

```
SOURCE_ROLE:  
direct evidence  
path lead  
version neighbor  
draft artifact  
counterexample  
unknown
```

A path lead cannot be cited as evidence until checked.

Step 5: Check version status

```
VERSION_STATUS:  
current  
superseded  
duplicate  
near duplicate  
excerpt  
unknown
```

Latest does not always mean valid. Older does not always mean obsolete. The rule is content status before filename confidence.

Step 6: Compare object-in and object-out

```
OBJECT_IN:  
what arrived
```

```
OBJECT_OUT:  
what the answer returns
```

```
MATCH:  
same, narrowed, expanded, substituted, unknown
```

If the object has been substituted, the answer should stop and repair route.

Step 7: State unknowns before action

```
UNKNOWN:  
missing source  
missing page  
missing version  
missing user permission  
missing test
```

Unknown is a status, not a failure of style.

7. Audit Table

Field	Question	Allowed output
Object	What is the live object?	named object
Terms	What governs this turn?	constraints, vetoes, format
Old film	What archive material is nearby?	paths, source families
Lamp	What may steer now?	current terms, chosen path
Source role	What is the source allowed to do?	evidence, lead, draft, counterexample
Version status	What is the version relation?	current, duplicate, superseded, unknown
Object match	Did the object survive transformation?	same, narrowed, expanded, substituted
Next move	What happens now?	answer, ask, hold, fetch, repair

8. Metrics

The following metrics can be used in tests.

8.1 Fetch Burden

`FETCH_BURDEN =`
how often the user must provide exact filename, path, quote, or section

A lower fetch burden is useful only if source substitution does not rise.

8.2 False Source Pull

`FALSE_SOURCE_PULL =`
how often the system retrieves a plausible but wrong source family

This is measured by source adjudication after retrieval.

8.3 Lamp Survival

`LAMP_SURVIVAL =`
whether current user terms survive after archive retrieval

A failure occurs when an archive habit overrides current constraints.

8.4 Object Survival

`OBJECT_SURVIVAL =`
whether the object returned by the answer is the same object under work

Allowed transformations must be named: narrowed, expanded, merged, split, or translated.

8.5 Source Role Accuracy

SOURCE_ROLE_ACCURACY =
whether the system labels a source as evidence, lead, draft, or unknown in line
with inspection

8.6 Version Discipline

VERSION_DISCIPLINE =
whether duplicates, older drafts, and newer copies are separated by content
status

9. Test Design

A paired test can compare two archive assistants loaded with the same corpus.

Task form

The user gives an object cue without path:

"Find the old-film repair."
"Use the source-near is not source-true law."
"Return the paper about object memory not path memory."

Measured outputs

1. Did the assistant reach the intended source family?
2. Did it ask for the exact path when the object cue was enough?
3. Did it label source role before using content?
4. Did it preserve the current user terms?
5. Did it mark uncertainty when route evidence was partial?
6. Did it invent a source edge?

Required comparison

The same prompt must be run against:

same corpus, no custody compression
same corpus, custody compression

The claim is not that one system is globally superior. The test only measures whether custody compression reduces fetch burden while containing false source pull.

10. What the Model Adds

Old Film, Live Lamp adds four design commitments.

10.1 Memory role before memory use

A remembered item must be assigned a role before it enters an answer.

10.2 Active state above archive mass

Current user terms steer over older material unless the user asks to restore an older route.

10.3 Source edge above thematic fit

A source that fits is not evidence until inspected.

10.4 Object survival as output test

The answer is tested against the object, not against fluency.

11. Limits

This paper does not claim a general benchmark result. It reports a working model derived from one large research archive and turns it into a testable protocol. The method still depends on source extraction quality, filename hygiene, user feedback, context-window behavior, and model reliability. It also requires honest labels for unknowns, partial reads, duplicate bodies, and version forks.

The model does not solve retrieval. It governs retrieval after it occurs. It does not make summaries true. It marks the role a summary may play. It does not remove the need for human judgment. It gives human judgment a visible surface.

12. Conclusion

Long-horizon AI memory should not be measured only by how much past material can be stored or retrieved. The deeper question is custody: what remembered material is allowed to do in the present turn.

Old Film, Live Lamp separates archive from steering surface. Old film keeps evidence, paths, drafts, and prior outputs. The live lamp holds current object, current terms, current permission, and current next move. Rolling

state moves the lamp without letting the archive become driver by mass.

The keeper is:

The archive remembers the shelf.
The lamp protects the object.

Internal Corpus Sources

[C1] extra_extra_expanded/old_film_live_lamp_object_custody_long_horizon_ai_memory_public_v0_007.md [C2]
extra_extra_expanded/source_nearness_under_12a_compression_v0_001.md [C3]
extra_extra_expanded/new_paper_gap_object_memory_is_not_path_memory_seed_v0_001.md [C4]
extra_extra_expanded/pointerless_archive_retrieval_12a_compression_paper_v0_002.md [C5]
extra_extra_intake_report.md [C6] extra_extra_manifest.csv [C7]
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Appendix A: Minimal Custody Card

OBJECT:
TERMS:
OLD_FILM:
LIVE_LAMP:
SOURCE_ROLE:
VERSION_STATUS:
OBJECT_IN:
OBJECT_OUT:
UNKNOWN:
NEXT_MOVE:

Appendix B: Source Role Ladder

DIRECT_EVIDENCE:
content was inspected and supports the claim

PATH_LEAD:
path or title suggests where to inspect

VERSION_NEIGHBOR:
nearby version may help but is not the target yet

DRAFT_ARTIFACT:
generated or working text, useful but not proof

COUNTEREXAMPLE:
source contradicts or limits the proposed claim

UNKNOWN:
source edge is absent or unread